

CROSSCERT

Web-Based Event Management & Certificate Automation Platform. Built for Holy Cross of Davao College — Office of the Vice President for Academic Affairs.

📅

March 2026

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Norman's Model · ACM Framework

👤

Admin + Participant Guide

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01

System Overview

Platform capabilities and feature summary

CROSSCERT is a next-generation, web-based platform that digitizes the full lifecycle of institutional events — from creation and registration, to attendance tracking and automated certificate delivery. Built for **Holy Cross of Davao College**, it serves two core user roles: Admins who manage events, and Participants who register and receive certificates.

FEATURE	WHO USES IT	WHAT IT DOES
Event Management	Admin	Create, manage, and publish events with full status control
Digital Tickets	Participant	Receive encrypted QR access passes upon registration
QR Attendance	Admin	Scan participant QR codes for real-time check-in and check-out
Evaluations	Participant	Submit post-event feedback to trigger certificate release
Certificates	Both	Auto-generate and deliver hi-res PDF certificates via email
Insights	Admin	View attendance, engagement, and participation analytics
Dark / Light Mode	Both	Toggle theme for your preferred display environment

Technology Stack

Frontend

Next.js 15 · TypeScript · Tailwind CSS · Radix UI

Database

PostgreSQL / SQLite

Backend

Django 5.1 · Python · ReportLab PDF

Messaging

Automated SMTP Email Delivery

02

Interaction Frameworks

Norman's Model and the ACM HCI Framework

Norman's 7-Stage Model of Interaction

Every feature in CROSSCERT is mapped to Donald Norman's seven-stage cycle, so users always know what to do, what is happening, and what to expect from the system at every point.

1

Goal

What do I want to accomplish?

2

Plan

What are my options?

3

Specify

What specific action do I take?

4

Perform

I click, type, or scan.

5

Perceive

What does the system show me?

6

Interpret

What does the feedback mean?

7

Compare

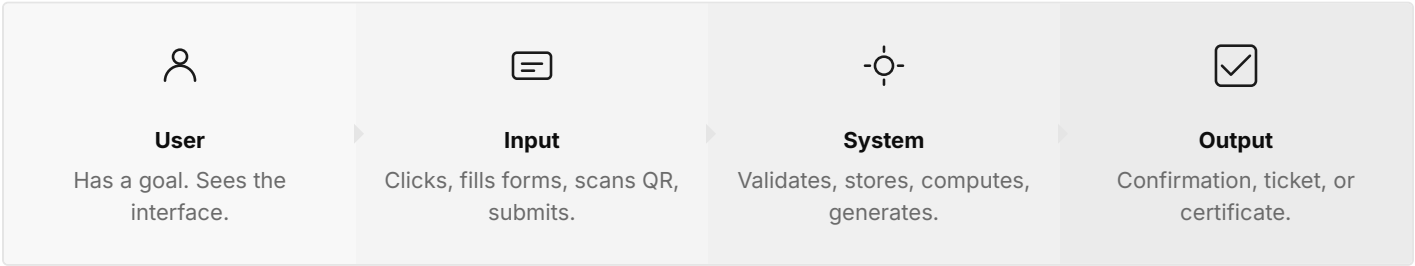
Did the system do what I intended?

Applied Example — Registering for an Event

STAGE	USER EXPERIENCE IN CROSSCERT
1 Goal	I want to attend the upcoming seminar.
2 Plan	Navigate to Events list, find the seminar.
3 Specify	Click the event card to open its landing page.
4 Perform	Click the Register Now button.
5 Perceive	A success modal appears; QR ticket delivered to email.
6 Interpret	My registration is confirmed.
7 Compare	QR ticket is visible in My Events — goal achieved.

ACM Framework — User · Input · System · Output

The ACM HCI Framework models interaction as a continuous loop. Every action in CROSSCERT follows this four-stage cycle.



03

User Roles

Admin and Participant capabilities

 **Admin**

✓ Create & manage events

✓ Control event lifecycle

✓ Scan QR for attendance

✓ Review evaluations

✓ Generate certificates

✓ View analytics & insights

✓ Manage system settings

 **Participant**

✓ Discover & browse events

✓ Register for events

✓ Receive digital QR ticket

✓ Submit post-event evaluation

✓ Download PDF certificate

✓ Manage bookmarks

✓ View personal settings

04

Core Interaction Flows

User — Input — System — Output across all primary journeys

Three primary flows drive the CROSSCERT experience, documented using the ACM User-Input-System-Output model.

Flow A — Event Creation Lifecycle (Admin)

STEP	USER INPUT	SYSTEM PROCESS	OUTPUT
Create	Click "Create Event"	Validates fields, stores record	Event saved as Draft
Publish	Change status to Scheduled	Updates state, notifies users	Event visible to participants
Go Live	Change status to Live	Enables QR scanner module	Check-in module active
Conclude	Change status to Concluded	Closes registration; opens evaluations	Evaluation forms sent

Flow B — Participant Registration & Check-In

STEP	USER INPUT	SYSTEM PROCESS	OUTPUT
Discover	Browse events list	Fetches all Scheduled events	Event cards displayed
Register	Click "Register Now"	Validates eligibility, slots; creates ticket; generates QR	Success modal; QR sent to email
Check-In	Show QR to admin scanner	Decodes QR, logs timestamp	Checked In — recorded
After Event	—	Admin sets Concluded status	Evaluation prompt appears

Flow C — Evaluation & Certificate Issuance

STEP	USER INPUT	SYSTEM PROCESS	OUTPUT
Evaluate	Submit evaluation form	Stores response; marks participant as evaluated	Submission confirmed
Generate	Admin clicks "Generate Certificates"	Checks eval status; runs ReportLab PDF engine	Hi-res PDF built per participant
Deliver	—	SMTP sends PDF to each participant; saves to portfolio	Email sent + download available

Important: Certificates are not generated until the participant has submitted their evaluation. This is required to trigger the PDF generation process.

05

System Status Messages

All platform feedback states and their meaning

STATUS	MEANING	WHAT TO DO
Registration Successful	You are registered	Check your email for the QR ticket
Check-In Recorded	Attendance captured	You may now enter the event
Evaluation Submitted	Feedback saved	Wait for certificate generation
Certificate Ready	PDF generated	Go to /participant/certificates to download
Already Registered	Duplicate registration detected	Check My Events for your existing ticket
Event Full	No slots remaining	Bookmark for future announcements
Invalid QR Code	Code unrecognized	Ensure you are showing the correct event ticket
Evaluation Required	Certificate blocked	Submit your evaluation first

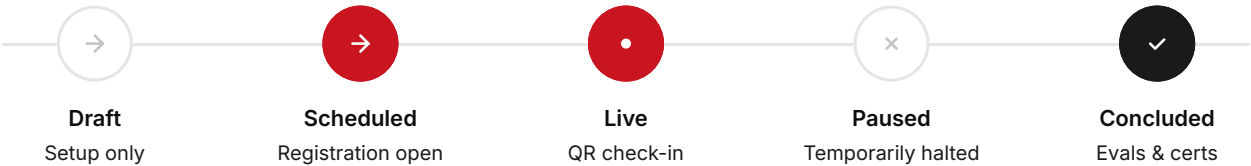
06

Event Lifecycle

State-driven event management

Event Status Progression

Every event passes through a defined sequence of statuses. Each status controls what participants and admins can do.



STATUS	MEANING	PARTICIPANT CAN REGISTER?
Draft	Event is being set up	No
Scheduled	Event is published and upcoming	Yes
Live	Event is currently happening	QR check-in active
Paused	Temporarily suspended	No new check-ins
Concluded	Event has ended	Evaluations open

07

UI Design & Visual Language

Color system, typography, animation, and component patterns

Design Philosophy

PRINCIPLE	HOW IT MANIFESTS
Clarity	Minimal clutter — every element on screen has a purpose
Consistency	Same color, spacing, and interaction patterns across all pages
Feedback	Every user action produces immediate, visible confirmation
Delight	Smooth animations and glassmorphism effects make the UI feel alive

Color System — Brand Primary: Institutional Red

Light Mode

hsl(354, 86%, 40%) · #B80B1C

Dark Mode

hsl(0, 100%, 50%) · #FF0000

Gradient form: from-red-600 to-rose-600

Department Color Coding — Glassmorphism Badges

● STE

● CET

● SBME

● CHATME

● HUSOCOM

● COME

● CCJE

All badges use `backdrop-blur-md` for the frosted glass effect throughout event cards.

Typography

Syne

DISPLAY FONT · WEIGHT 400-800

Headings, hero text, display titles, sidebar navigation labels.

Space Grotesk

BODY FONT · WEIGHT 300-700

Body text, labels, navigation, forms, and all readable content.

Animation System

ANIMATION	DESCRIPTION & USAGE
float	Gentle vertical oscillation (0 to -20px). Used on hero visual elements.
blob	Organic scale and translate morphing. Used on background glow decorations.
scan	Top to bottom sweep with opacity fade. Used on QR scanner overlay.
marqueeSlide	Infinite left scroll. Used on the department badge ticker strip on the landing page.
Framer Motion	Splash entrance: scale 0.8 to 1, opacity 0 to 1. Exit: -y100 opacity 0 over 0.8s. Easing [0.76, 0, 0.24, 1].

3D Lanyard: The landing page features a physics-based 3D lanyard built with React Three Fiber and @react-three/rapier. The CROSSCERT ID card is draggable in real-time.

Iconography — Lucide React

ICON	USED FOR	ICON	USED FOR
LayoutDashboard	Dashboard navigation	Calendar	Events section
Users	Participants section	QrCode	QR Attendance / Check-In
FileText	Certificates	BarChart3	Insights / Analytics
Bookmark	Bookmarks	Award	Participant Certificates
Sparkles	Role badge indicator	LogOut	Logout button

08 Participant Walkthrough

Step-by-step guide from login to certificate download

Step 1 — Log In to the

✓ Platform

Path: crosscert.hcdc.edu.ph

1 Navigate to the CROSSCERT URL

Open your browser and go to the platform URL. The animated splash screen and 3D lanyard loads on the landing page.

2 Click "Sign In" and enter your credentials

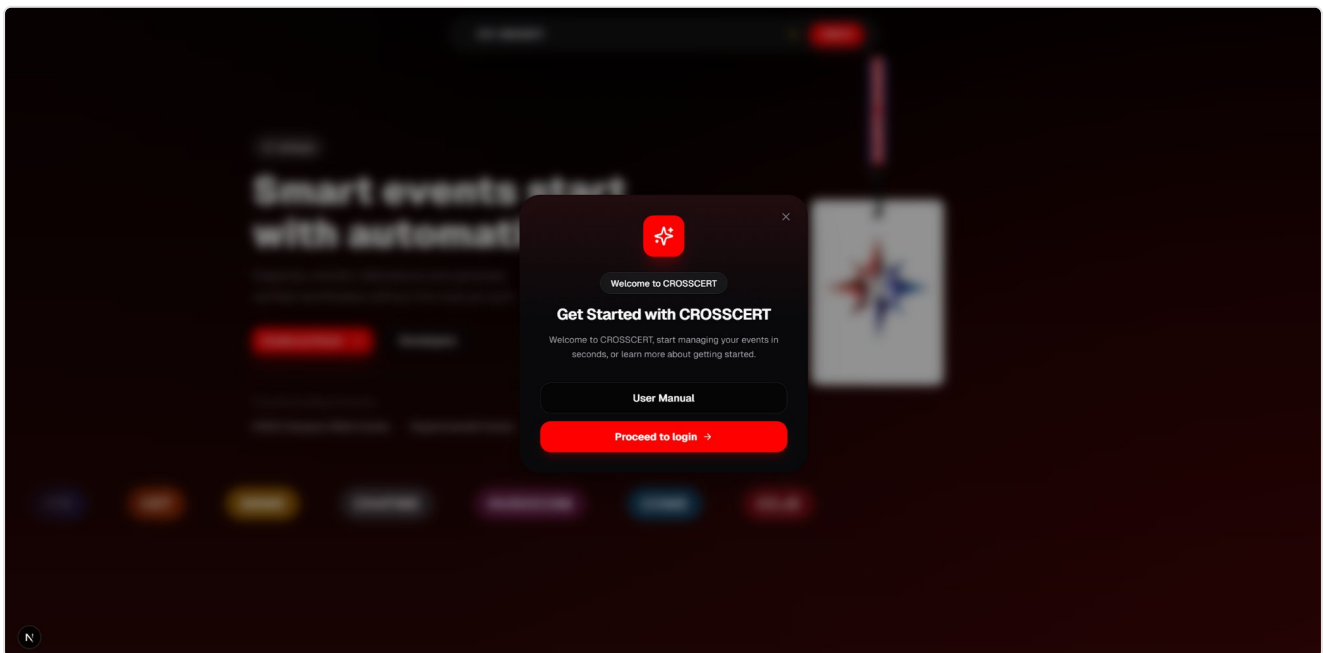
Enter your Email and Password, then click Sign In.

Output: You land on /participant/dashboard

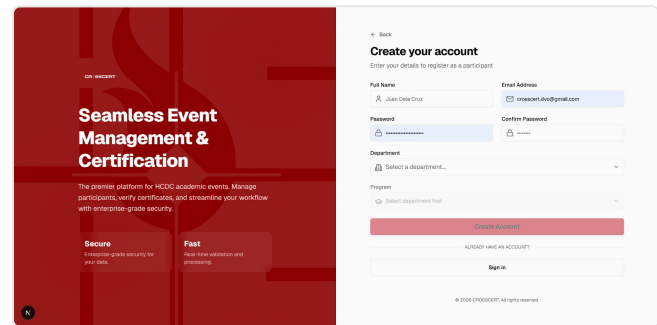
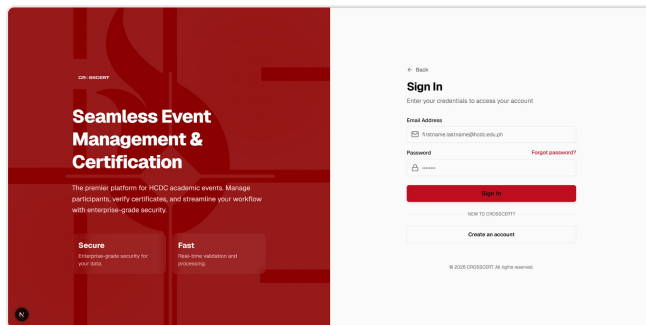
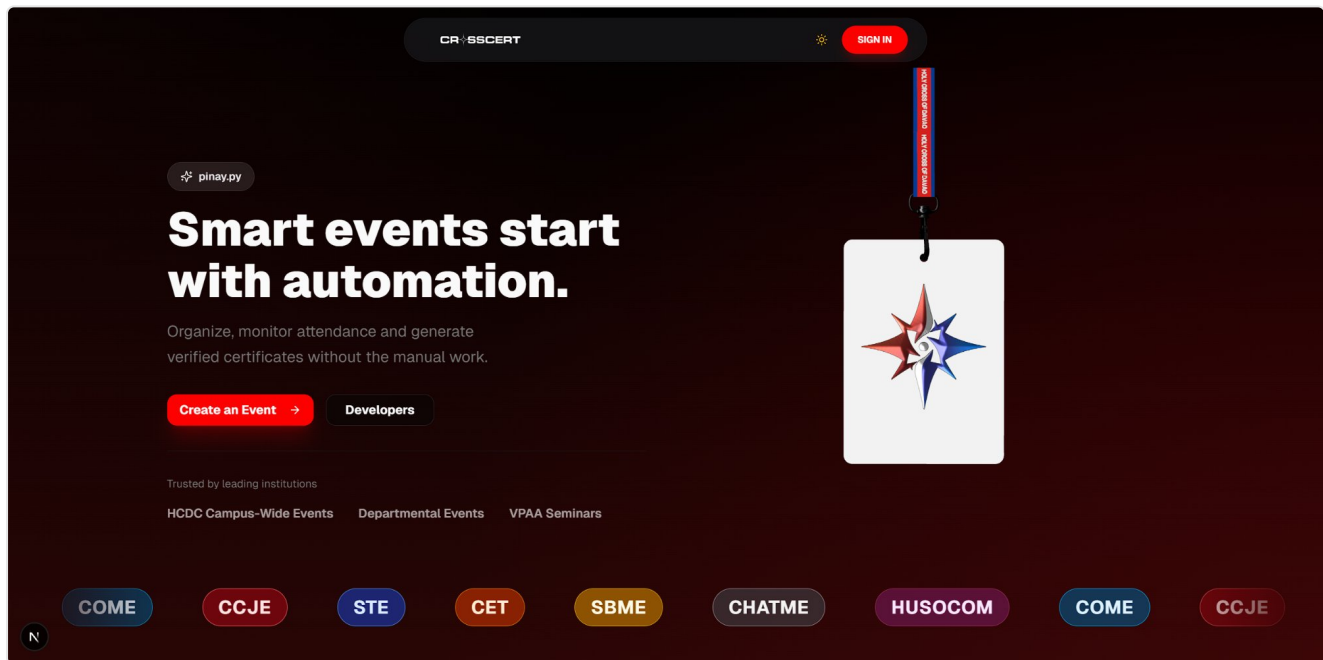
3 No account? Sign Up first

Click Sign Up, fill in Full Name, Email, Department, Password, and Confirm Password. Verify your email before returning to Sign In.

— WELCOME MODAL — USER MANUAL OR PROCEED TO LOGIN



First-time visitors see this modal — click "User Manual" to access this guide, or "Proceed to login" to go straight to the platform.



1

Review Upcoming Events

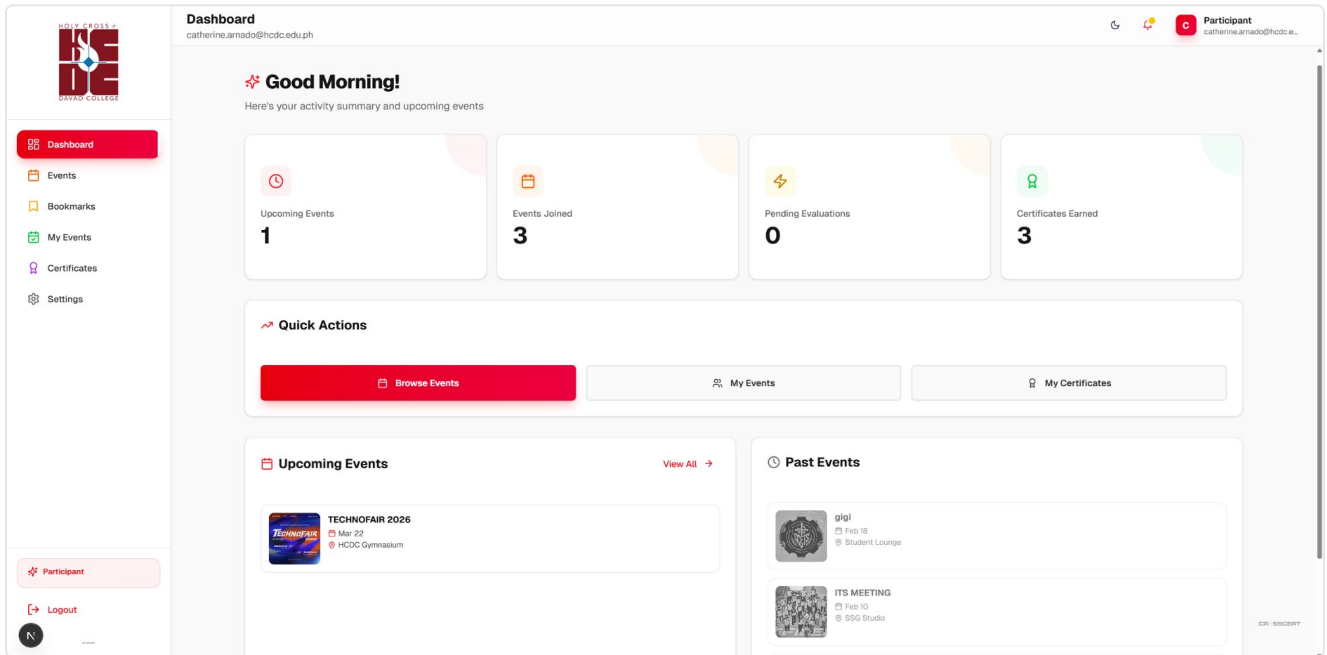
Events you are already registered for appear at the top with their current status.

2

Check Notifications and Certificates

New event alerts and certificate-ready messages from admins appear in the notification panel. Recently issued PDFs appear for quick access.

— PARTICIPANT DASHBOARD



The screenshot displays the Participant Dashboard for a user named Catherine Arnado. The interface includes a sidebar with navigation links: Dashboard, Events, Bookmarks, My Events, Certificates, and Settings. The main content area features a 'Good Morning!' greeting, a summary of activity and upcoming events, and four key metrics: Upcoming Events (1), Events Joined (3), Pending Evaluations (0), and Certificates Earned (3). Below these metrics is a 'Quick Actions' section with buttons for 'Browse Events', 'My Events', and 'My Certificates'. The 'Upcoming Events' section lists 'TECHNOFAIR 2026' on March 22 at HCDC Gymnasium. The 'Past Events' section lists 'gigi' on February 18 at Student Lounge and 'ITS MEETING' on February 10 at SSG Studio. The user's profile information, including the name 'Participant' and email 'catherine.arnado@hcdc.edu.ph', is visible in the top right corner.

Dashboard
catherine.arnado@hcdc.edu.ph

Good Morning!
Here's your activity summary and upcoming events

Upcoming Events
1

Events Joined
3

Pending Evaluations
0

Certificates Earned
3

Quick Actions

[Browse Events](#) [My Events](#) [My Certificates](#)

Upcoming Events [View All](#)

TECHNOFAIR 2026
Mar 22
HCDC Gymnasium

Past Events

gigi
Feb 18
Student Lounge

ITS MEETING
Feb 10
SSG Studio

Participant
Logout

OR: SIGN OUT

1 Click Events in the sidebar

A grid of event cards appears, each showing Event Name, Date and Time, Department badge, available slots, and a bookmark icon.

2 Search or Filter events

Use the Search bar to find a specific event by name. Filter by department or date if needed.

3 Open the Event Landing Page

Click any card to open the full event page: cover banner, description, date, location, slots remaining, and the Register button.

DISCOVER EVENTS

The screenshot shows the 'Discover Events' interface. On the left is a sidebar with navigation links: Dashboard, Events (highlighted), Bookmarks, My Events, Certificates, and Settings. Below these is a 'Participant' section with a 'Logout' button. The main content area is titled 'Events' with the user email 'catherine.amado@hcdc.edu.ph'. It features a 'Discover Events' section with a search bar and filters. The filters include 'School Year' (set to 'All Years') and 'Semester' (with 'All Semesters' selected). A horizontal menu shows departmental filters: ALL, HCDC EVENTS, CCJE, CET, CHATME, HUSOCOM, COME, SBME, and STE. Below this, there are buttons for 'Upcoming (1)' and 'Past Events (4)'. The main event card is for 'TECHNOFAIR 2026', dated 'MARCH 18 01:00 PM', at 'HCDC Gymnasium'. A 'Join Event' button is at the bottom of the card. On the right, a 'FILTER BY MONTH' dropdown menu is open, showing a list of months from JANUARY to DECEMBER, with 'All Months' selected at the top.

Step 4 — Register for an



Event

Core participant interaction

1 Review the event details and click "Register Now"

Confirm the date, location, and description. Click the red Register Now button.

S System validates your registration

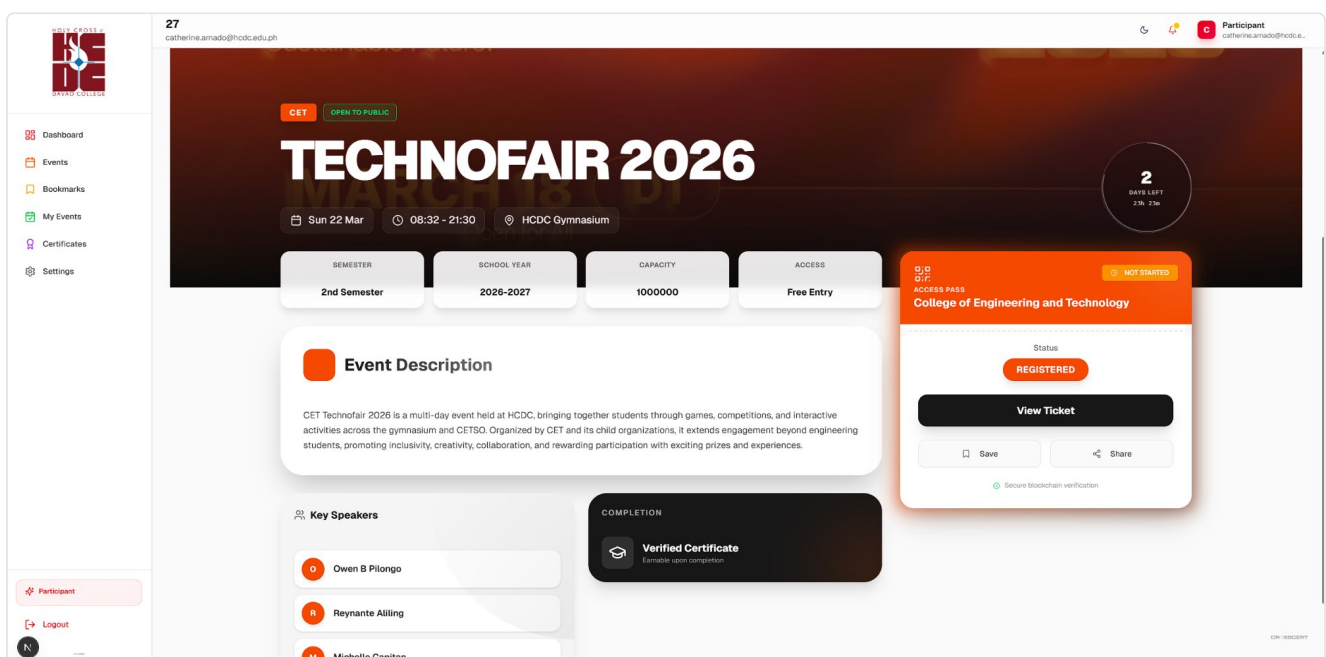
Checks: Are you signed in? Is the event Scheduled? Are there available slots? Are you not already registered? Then creates your registration record and generates an encrypted QR Digital Access Pass.

Output: Success modal + QR sent to your email

2 Event appears in My Events

The registered event is now visible in your My Events section with real-time status tracking.

— EVENT DETAILS & REGISTRATION



Event Full? If no slots remain, the Register button is disabled. Bookmark the event to watch for announcements.

Step 5 — Use Your QR Ticket at the Event

Digital Access Pass — QR Check-In

1 Find your QR ticket

Check your email inbox (or spam folder) for the QR pass. Or go to My Events in the app and tap your event to view it directly.

2 Present your QR at the entrance

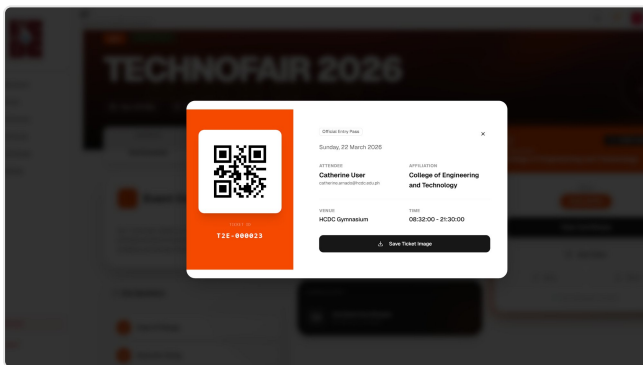
Maximize screen brightness. Hold the QR code flat and steady. Show it to the admin at the event entrance.

S Admin scans with the CROSSCERT scanner

The system decodes your QR, validates your registration, and logs the timestamp.

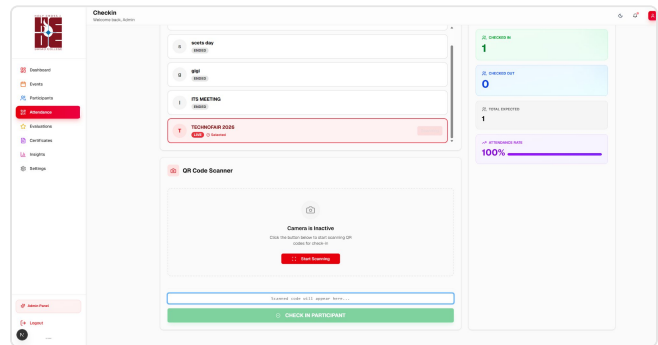
Output: Check-In Successful — attendance recorded

DIGITAL ACCESS PASS (QR TICKET)



Official Entry Pass — scannable at entrance

ADMIN QR CHECK-IN SCANNER



Admin scans participant QR to record attendance

Step 6 — Submit Your

≡ Evaluation

Required before certificate issuance

1 Go to My Events after the event concludes

Find the concluded event. An Evaluate button appears next to it. The event progress tracker shows REG → IN → OUT → EVAL stages.

2 Complete the feedback form

Select a star rating (1–5) across multiple categories: Content Quality, Instructor/Resource Speaker, Facilities/Venue, and Overall Experience. Optionally write comments.

3 Click "Submit Evaluation"

Your response is saved and your account is marked as evaluated, unlocking certificate generation.

Output: Thank you — evaluation confirmed

— MY EVENTS — PROGRESS TRACKER (REG · IN · OUT · EVAL)



Dashboard

Events

Bookmarks

My Events

Certificates

Settings

My events
catherine.arnado@hcdc.edu.ph

← Back

My Events
Track your registrations, attendance, and feedback

Running & Upcoming (1)History (3)



EVALUATED

TECHNOFAIR 2026

🕒 08:32:00 📍 HCDC Gymnasium

✓ REG

—

✓ IN

—

✓ OUT

—

✓ EVAL

View Ticket

✓ Mark as Done

— EVALUATION REMINDER MODAL

**Pending Evaluations**
Your feedback helps us improve!

You have 1 event(s) waiting for your feedback!

TECHNOFAIR 2026
March 22, 2026

Evaluate Now

Remind Me Later

View My Events

— EVALUATION FORM

Back

**Event Evaluation**
Help us improve by sharing your experience

TECHNOFAIR 2026

New Label *

1-5, 1st Mar, 2nd Mar, 3rd Mar, 4th Mar

Rate Your Experience

Content Quality

★★★★☆ 4/5

Instructor/Resource Speaker

★★★★☆ 4/5

Facilities/Venue

★★★★☆ 4/5

Overall Experience

★★★★☆ 4/5

1 Wait for admin to generate certificates

After all evaluations are collected, the admin triggers PDF generation. You receive an email notification when ready — including Registration Confirmation, Attendance Verified, Evaluate Reminder, and Certificate emails.

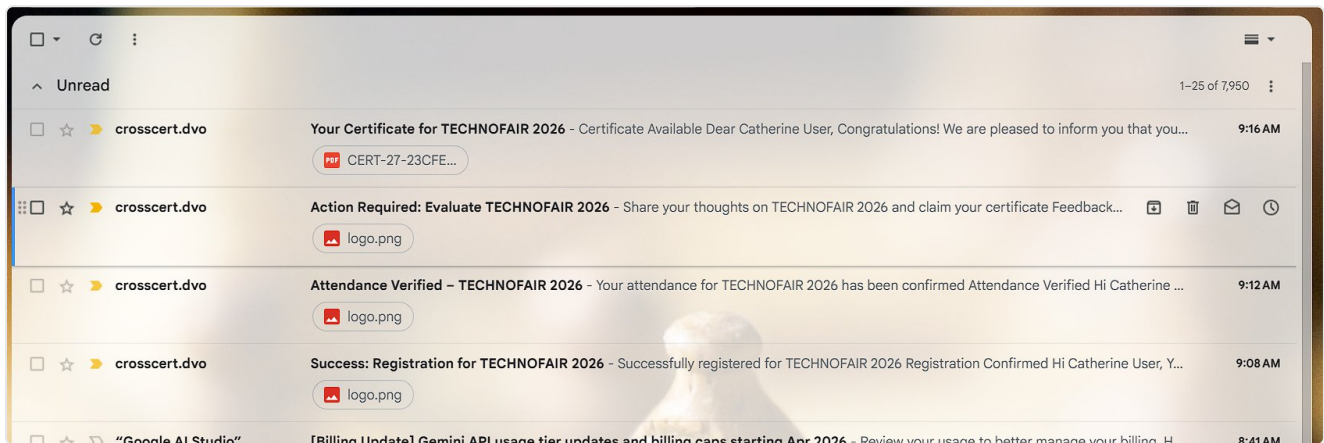
2 Go to Certificates in the sidebar

All your issued certificates appear as downloadable cards organized by event and date under "Access Granted."

3 Click "Download PDF"

A high-resolution, print-ready certificate opens or saves. It contains your Full Name, Event Title, Date, HCDC Institutional Header and Seal, Authorized Signatories, and a Unique Certificate ID.

Output: Print-ready PDF downloaded

EMAIL NOTIFICATIONS — FULL FLOW

Registration Confirmation · Attendance Verified · Evaluate Reminder · Certificate Ready



HOLY CROSS of
DAVAO COLLEGE

Certificates
catherine.arnado@hcdc.edu.ph

← Back

Access Granted

View and download your earned certificates and credentials



#CERT-27-...

TECHNOFAIR 2026
ISSUED ON MARCH 19, 2026

This certifies that the recipient has successfully completed the requirements for the above event.

PDF



ITS MEETING
ISSUED ON FEBRUARY

This certifies that the recipient has successfully completed the requirements for the above event.

PDF

Dashboard

Events

Bookmarks

My Events

Certificates

Settings

— CERTIFICATE OF PARTICIPATION — PDF OUTPUT



COLLEGE OF ENGINEERING AND TECHNOLOGY



CERTIFICATE OF PARTICIPATION

proudly awarded to

CATHERINE USER

In recognition for successfully participating in the _____ held on _____ organized by the Office of the Vice President for Academic Affairs (VPAA)

TECHNOFAIR 2026
March 22, 2026

Teresa P. Fabiana
VPAA Office

Owen B. Pilongo
Seminar Organizer

CROSSCERT

Auto-generated hi-res PDF certificate with HCDC branding, CROSSCERT seal, and authorized signatories

09 Admin Walkthrough

Step-by-step guide from event creation to certificate generation



Admin Step 1 — Log In as Admin

Role-based login and dashboard access

1

Navigate to CROSSCERT and click Sign In

Enter your Admin Email and Password, then click Sign In.

Output: /admin/dashboard with the Admin Sidebar

2

Admin Sidebar Navigation

Access: Dashboard, Events, Participants, Attendance (QR), Evaluations, Certificates, Insights, and Settings.



Admin Step 2 — View Admin

Dashboard

Path: /admin/dashboard

1

Review summary cards

At a glance: Active Events, Registered Participants, Checked In Today, Certificates Issued, and Evaluations Submitted.

2

Upcoming events and real-time check-in activity

Upcoming events with their current statuses are listed below summary cards. Check-in activity updates in real-time.



Admin Step 3 — Create an

Event

Path: /admin/events

1

Click Events in the sidebar, then "+ Create Event"

The event creation form opens.

2

Fill in all required fields

Event Name · Description · Department · Date and Time · Location · Cover Image · Theme Color · Maximum Slots.

S

Click "Create Event" — system validates and stores

Output: Event saved as Draft — not visible to participants yet

Admin Step 4 — Manage Event

Status

Advance through the lifecycle

1 Open the event and find the Status control

Go to the Events list, click your event, then use the Status dropdown or button.

2 Advance through statuses as needed

Draft → Scheduled: event publishes and registration opens. Scheduled → Live: QR scanner activates. Live → Concluded: evaluation forms sent to all attendees.

Status changes are instant — visible to participants live

Admin Step 5 — Edit or Delete an

Event

Path: /admin/events

1 To Edit: click the pencil icon on the event card

Modify any field, then click Save Changes.

Output: Toast — Event updated successfully

2 To Delete: click the trash icon and confirm

A confirmation dialog appears. Click Confirm Delete. The event and all associated data are permanently removed. This action cannot be undone.

Admin Step 6 — Run QR Scanner (Check-

In)

Path: /admin/checkin · Event must be Live

1 Click Attendance, select the Live event, click "Start Scanner"

Your camera activates. Point it at the participant's QR ticket.

System decodes QR and logs attendance

Success: Check-In Recorded — timestamp logged automatically

Error: Already Checked In / Invalid QR / Wrong Event

2 Check-Out at end of event

Scan the same QR as participant exits. The system records the check-out timestamp and calculates attendance duration. A participant cannot check out before checking in.



Admin Step 7 — Review Evaluations

Path: /admin/evaluations · Event must be Concluded

1

Click Evaluations and select the event

All submitted evaluations appear with star ratings (1–5), written comments, participant name, department, and submission timestamp.

2

Review and use for improvement

Use ratings and comments to assess event quality and inform future event planning.

Note: Only participants who submitted an evaluation will receive a certificate.

Admin Step 8 — Generate



Certificates

Path: /admin/certificates

1

Click Certificates, select the Concluded event

The eligible participant list appears. Participants with a submitted evaluation are marked Ready. Those without are marked Blocked.

2

Click "Generate Certificates"

ReportLab builds one certificate per eligible participant, applying event branding, name, date, and HCDC seal.

S

System delivers and stores all certificates

Each PDF is emailed via SMTP to the participant's inbox and saved to their digital portfolio.

Output: Certificates issued to X participants

Admin Step 9 — Configure



Settings

Path: /admin/settings

SECTION	WHAT YOU CAN CONFIGURE
Theme	Toggle Light / Dark mode platform-wide
Account Security	Change admin password
Email Configuration	Set SMTP host, port, sender address for QR tickets and certificates
Institutional Branding	Upload updated HCDC logos and adjust platform color scheme

10

Quick Reference Map

Complete feature map for both roles

- 01 Log In as Admin — /admin/dashboard
- 02 View Admin Dashboard — summary cards and events
- 03 Create a New Event — fill form, save as Draft
- 04 Publish Event — change status to Scheduled
- 05 Go Live — activate QR scanner module
- 06 Edit or Delete Event — pencil or trash icon
- 07 Manage Participants — search, filter, view profiles
- 08 Run QR Scanner — check-in and check-out
- 09 Review Evaluations — after Concluded status
- 10 Generate Certificates — for evaluated participants
- 11 View Insights — /admin/insights analytics
- 12 Configure Settings — SMTP, branding, password

- 01 Sign Up or Log In — /participant/dashboard
- 02 View Dashboard — events, notifications, certs
- 03 Browse Events — /participant/events or /discover
- 04 Register for Event — click Register Now
- 05 Get QR Ticket — via email or My Events
- 06 Show QR at Entrance — admin scans the code
- 07 Check My Events — track attendance and status
- 08 Bookmark Events — save events for later
- 09 Submit Evaluation — after event concludes
- 10 Download Certificate — /participant/certificates
- 11 Manage Settings — profile, password, theme
- 12 Log Out — bottom of the sidebar